Opgavesæt 2 i vektoranalyse

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1 Exercise 1

Let F be a vector field on $\mathbb{R}^3$ given by:

$$F(x,y,z) = (xy^2z, x^2yz, \frac{1}{2}x^2y^2)$$

calculate the integral:

$$\oint_C \mathbb{F} \cdot dr$$

where $C = \{(x,y,z) \in \mathbb{R}^3 | x^2 + y^2 = 1, z = 4\}$ (for positive orientation in the xy-plane)

Because x and y is defined as a unit circle in C, with a constant Z value, this can be rewritte as a parameterization of the unit circle.

$$x = cos(\theta), \qquad y = sin(\theta), \qquad z = 4$$

This changes the bound of integration to:

$$\int_0^{2\pi}$$

Because of this change, a position vector is also created

$$r(\theta) = \begin{pmatrix} cos(\theta) \\ sin(\theta) \\ 4 \end{pmatrix}$$

In order to calculate the integral, $r(\theta)$ must be differentiated. This reflects the change in direction due to any changes of position along the curve.

$$r'(\theta) = \begin{pmatrix} -sin(\theta) \\ cos(\theta) \\ 0 \end{pmatrix}$$

Multiplying $r'(\theta)$ by any infitisimal change, yields the displacement vector:

$$dr = r'(\theta)\ d\theta$$

The position vector, can then be plugged into the equation for the vector field, yielding: $F(r(\theta))$

$$F(r(\theta)) = \begin{pmatrix} 4cos(\theta)sin^2(\theta) \\ 4sin(\theta)cos(\theta) \\ \frac{1}{2}cos^2(\theta)sin^2(\theta) \end{pmatrix}$$

This makes the line integral

$$\int_0^{2\pi} \mathbb{F} \cdot dr \qquad = \int_0^{2\pi} F(r(\theta)) \cdot r'(\theta)\ d\theta \qquad = \qquad \begin{pmatrix} 4cos(\theta)sin^2(\theta) \\ 4sin(\theta)cos(\theta) \\ \frac{1}{2}cos^2(\theta)sin^2(\theta) \end{pmatrix} \cdot \begin{pmatrix} -sin(\theta) \\ cos(\theta) \\ 0 \end{pmatrix}\ d\theta$$

Yielding two terms

$$\int_0^{2\pi} -(4cos(\theta)sin^3(\theta)) + (4cos^3(\theta)sin(\theta))d\theta$$

This will be split up into two integrals and then solved
First term

$$\int -(4cos(\theta)sin^3(\theta))d\theta$$

$$\frac{d}{d\theta}sin(\theta) = cos(\theta), \qquad u = sin(\theta), \qquad \frac{du}{d\theta} = cos(\theta), \qquad du = cos(\theta)$$

$$-4\int cos(\theta)d\theta sin^3(\theta) \Rightarrow -4\int u^3 du \Rightarrow -4\frac{u^4}{4} + C = -sin^4(\theta) + C$$

Second Term

$$\int (4cos^3(\theta)sin(\theta))d\theta$$

$$\frac{d}{d\theta}cos(\theta) = -sin(\theta), \qquad u = cos(\theta), \qquad du = -sin(\theta), \qquad sin(\theta)d\theta = -du$$

$$4\int cos^3(\theta)\ sin(\theta)\ d\theta \ \Rightarrow 4\int u^3(-du) \Rightarrow -4\frac{u^4}{4} + C = -cos^4(\theta) + C$$

$$\left[-sin^4(\theta) - cos^4(\theta)\right]_0^{2\pi} \Rightarrow (-sin^4(2\pi) - cos^4(2\pi)) - (-sin^4(0) - cos^4(0)) = (-0^4 - 1^4) - (-0^4 - 1^4) = (-1) - (-1) = 0$$

2 Exercise 2

Let F be a vector field on $\mathbb{R}^3$ given by:

$$F(x, y, z) = (3x, 2y, -z)$$

- *Calculate $\nabla \cdot F$ and $\nabla \times F$ and find all function ϕ such that $\nabla\phi = F$*
- *Calculate the flux of F out of the unit sphere $S^2 = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 + z^2 = 1\}$*

The product of ∇ and F, which is vector field, yields a scalar field, this can be found as:

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x}(3x) + \frac{\partial F_y}{\partial y}(2y) + \frac{\partial F_z}{\partial z}(-z) = 3 + 2 - 1 = 4$$

This is also known as the divergence, this tells us something about the flux of the vector field, leaving a small closed surface. The curl of the vector field tells us about local rotation and can calculated as follows:

$$\nabla \times \mathbf{F} = \begin{pmatrix} \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\partial(-z)}{\partial y} - \frac{\partial(2y)}{\partial z} \\ \frac{\partial(3x)}{\partial z} - \frac{\partial(-z)}{\partial x} \\ \frac{\partial(2y)}{\partial x} - \frac{\partial(3x)}{\partial y} \end{pmatrix} = (0 - 0, 0 - 0, 0 - 0) = 0$$

This tells us that the vector field is *Irrotational* and also convervative. This also means that the field is defined everywhere in the space. If a vector field **F** is convervative, then the corresponding scalar field ϕ is called the potential for **F**. The gradient consists of three partial derivates that must yield the vector field itself:

$$\frac{\partial \phi}{\partial x} = 3x \qquad \frac{\partial \phi}{\partial y} = 2y \qquad \frac{\partial \phi}{\partial z} = -z$$

Integrate the first term with respect to x, however we need to introduce arbitrary functions g(y,z) and h(z) during each integration, to recover any lost information, as we cannot be sure if the original function contained any terms relating to y and z. Because:

$$\nabla\phi = \begin{pmatrix} \frac{\partial \phi}{\partial x} \\ \frac{\partial \phi}{\partial y} \\ \frac{\partial \phi}{\partial z} \end{pmatrix}, \qquad \textit{and} \qquad \nabla\phi = \mathbf{F} \qquad \Rightarrow \qquad \mathbf{F} = \begin{pmatrix} \frac{\partial \phi}{\partial x} \\ \frac{\partial \phi}{\partial y} \\ \frac{\partial \phi}{\partial z} \end{pmatrix}$$

This means that we must find some scalar function ϕ, that when partial differentiated yields the vector field. As listed above, the three partials, we can start by integration the x component.

$$\int 3x \; dx \Rightarrow \phi = \frac{3x^2}{2} + g(y, z)$$

If we differentiate ϕ for y, the x term goes away and we're left with:

$$\frac{\partial \phi}{\partial y} = \frac{\partial}{\partial y}\left(\frac{3}{2}x^2 + g(y, z)\right) \Rightarrow \frac{\partial \phi}{\partial y} = \frac{\partial g(y, z)}{\partial y} = 2y$$

And this partial derivate is integrated:

$$\int \frac{\partial g(y, z)}{\partial y} \; dy = y^2 + h(z) \Rightarrow \phi = \frac{3x^2}{2} + y^2 + h(z)$$

Same process as before, if we differentiate ϕ for z, the x and y terms goes away, and we're left with:

$$\frac{\partial \phi}{\partial z} = \frac{\partial}{\partial z}\left(\frac{3}{2}x^2 + y^2 + h(z)\right) \Rightarrow \frac{\partial \phi}{\partial z} = \frac{\partial h(z)}{\partial z} = -z$$

Integrating this, becomes:

$$\int \frac{\partial h(z)}{\partial z} = -\frac{z^2}{2} + C$$

And combining all the terms, yields the final expression:

$$\phi = \frac{3x^2}{2} + y^2 - \frac{z^2}{2} + C$$

To find the flux flowing out of the unit sphere, the divergence theorem can be used. This relates the flux throught the surface to the volume integral of the divergence. Here we transform the surface S into a solid sphere or ball B, but this also requires a change to spherical coordinates: (r, θ, ϕ)

$$\iint_{S^2} \mathbf{F} \cdot \mathbf{n} \; dS = \iiint_B \nabla \cdot \mathbf{F} \; dV$$

from ealier, the divergence was calculated:

$$\nabla \cdot \mathbf{F} = 3 + 2 - 1 = 4$$

This yields the integral:

$$\iiint_B 4 \; dV = \int_0^1 \int_0^\pi \int_0^{2\pi} 4r^2 sin(\phi) \; d\theta \; d\phi \; dr$$

Where $dV = r^2 sin(\phi) \; d\theta \; d\phi \; dr$, where the bounds of integration are defined as:

$$0 \leq r \leq 1, \qquad 0 \leq \phi \leq \pi, \qquad 0 \leq \theta \leq 2\pi$$

First integral:

$$= \int_0^{2\pi} 4r^2 sin(\phi) \; d\theta \Rightarrow 4r^2 sin(\phi) \int_0^{2\pi} 1 \; d\theta \Rightarrow 4r^2 sin(\phi) \left[\theta\right]_0^{2\pi} = 4r^2 sin(\phi)(2\pi - 0) = 8\pi r^2 sin(\phi)$$

Second integral:

$$= \int_0^{\pi} 8\pi r^2 sin(\phi) \; d\phi \Rightarrow 8\pi r^2 \int_0^{\pi} sin(\phi) \; d\phi \Rightarrow 8\pi r^2 \left[-cos\right]_0^{\pi}$$

$$\Rightarrow 8\pi r^2((-cos(\pi)) - (-cos(0))) = 8\pi r^2((-(-1)) - (-1)) = 16\pi r^2$$

Last and outer Integral

$$= \int_0^1 16\pi r^2 \; dr \Rightarrow 16\pi \int_0^1 r^2 \; dr \Rightarrow \left[\frac{16\pi}{3} r^3\right]_0^1 = \frac{16\pi}{3}$$

3 Exercise 3

Define the potential ϕ as:

$$\phi(x,y,z) = ln(x^2+y^2+z^2),\ where\ (x,y,z) \in \mathbb{R}^3\{(0,0,0)\}$$

- Calculate $\nabla\phi$
- Calculate the flux of $\nabla\phi$ out of the sphere: $S^2 = \{(x,y,z) \in \mathbb{R}^3 | x^2+y^2+z^2 = R^2\}$ where R is a positive real number.

The gradient is defined as the partial derivative of each of the variables.
In this case the potential ϕ is a logarithmic function, meaning it has to be treated as an inner and an outer function, using the chain rule. $f(x) = ln(g(x)) \Rightarrow \frac{d}{dx}f(x) \Rightarrow f'(x) = \frac{1}{g(x)} \cdot g(x)$

$$\frac{\partial\phi}{\partial x} = \frac{\partial}{\partial x} ln(x^2+y^2+z^2) = \frac{1}{x^2+y^2+z^2} \cdot (2x+0+0) = \frac{2x}{x^2+y^2+z^2}$$
$$\frac{\partial\phi}{\partial y} = \frac{\partial}{\partial y} ln(x^2+y^2+z^2) = \frac{1}{x^2+y^2+z^2} \cdot (0+2y+0) = \frac{2y}{x^2+y^2+z^2}$$
$$\frac{\partial\phi}{\partial z} = \frac{\partial}{\partial z} ln(x^2+y^2+z^2) = \frac{1}{x^2+y^2+z^2} \cdot (0+0+2z) = \frac{2z}{x^2+y^2+z^2}$$

Yielding the final result:

$$\nabla\phi = \left(\frac{2x}{x^2+y^2+z^2}, \frac{2y}{x^2+y^2+z^2}, \frac{2z}{x^2+y^2+z^2}\right)$$

To calculate the flux out of the unit sphere S^2, this yields another scalar field, which is $\nabla \cdot \nabla$, this is also known as the *laplacian* of ϕ. This is possible if the scalar field is twice differentiable, which this is. And then afterwards integrate over the volume enclosed by the sphere.

$$\nabla \cdot \nabla\phi = \nabla^2\phi = \frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial\phi}{\partial z}\right)$$

Deriving the laplacian:

$$\nabla^2\phi = \frac{\partial}{\partial x}\left(\frac{2x}{x^2+y^2+z^2}\right) + \frac{\partial}{\partial y}\left(\frac{2y}{x^2+y^2+z^2}\right) + \frac{\partial}{\partial z}\left(\frac{2z}{x^2+y^2+z^2}\right)$$

This is evalutated using the quotient rule:$(\frac{u}{v})' = \frac{u'v-uv'}{v^2}$

$$\frac{\partial}{\partial x}\left(\frac{2x}{x^2+y^2+z^2}\right) = \frac{2(x^2+y^2+z^2)-4x^2}{(x^2+y^2+z^2)^2}$$
$$\frac{\partial}{\partial y}\left(\frac{2y}{x^2+y^2+z^2}\right) = \frac{2(x^2+y^2+z^2)-4y^2}{(x^2+y^2+z^2)^2}$$
$$\frac{\partial}{\partial z}\left(\frac{2z}{x^2+y^2+z^2}\right) = \frac{2(x^2+y^2+z^2)-4z^2}{(x^2+y^2+z^2)^2}$$
$$= \frac{2(x^2+y^2+z^2)-4x^2}{(x^2+y^2+z^2)^2} + \frac{2(x^2+y^2+z^2)-4y^2}{(x^2+y^2+z^2)^2} + \frac{2(x^2+y^2+z^2)-4z^2}{(x^2+y^2+z^2)^2}$$
$$\nabla^2\phi = \frac{2(x^2+y^2+z^2)-4x^2+2(x^2+y^2+z^2)-4y^2+2(x^2+y^2+z^2)-4z^2}{(x^2+y^2+z^2)^2}$$
$$= \frac{6(x^2+y^2+z^2)-4(x^2+y^2+z^2)}{(x^2+y^2+z^2)^2}$$

And by the definition of the sphere, this can be expressed as:

$$\nabla^2\phi = \frac{2}{x^2+y^2+z^2} = \frac{2}{R^2}$$

Flux through the sphere can then be evalutated using Gauss's theorem, where the surface integral can be replaced with a volume integral:

$$\Phi = \iiint_B \frac{2}{R^2}\ dV$$

Using bounds of integration, where $dV = r^2\ sin(\phi)\ dr\ d\phi\ d\theta$ and from the hamiltonian, the denominator $R^2 = r^2$ and R is the is used as the integration bound.

$$\Phi = \int_0^{2\pi}\int_0^{\pi}\int_0^{R} 2sin(\theta)\ dr\ d\phi\ d\theta$$

First integral:

$$2sin(\phi)\int_0^R\ dr \Rightarrow 2sin(\phi)\,[r]_0^R = 2sin(\phi)\cdot(R-0) = 2R\ sin(\phi)$$

Second integral:

$$\int_0^{\pi} 2R\ sin(\phi)\ d\phi = 2R\int_0^{\pi} sin(\phi)\ d\phi \Rightarrow 2R\,[-cos(\phi)]_0^{\pi}$$

$$\Rightarrow 2R\cdot(-cos(\pi)-(-cos(0))) = 2R(-(-1)-(-1)) = 2R(1+1) = 4R$$

Third integral:

$$\int_0^{2\pi} 4R\ d\theta = 4R\int_0^{2\pi} d\theta \Rightarrow 4R\,[\theta]_0^{2\pi} \Rightarrow 4R(2\pi-0) = 8\pi R$$

This results in the flux being:

$$\Phi = 8\pi R$$

4 Exercise 4

Let $C = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 - 2x = 3\}$, (positive orientation) and let S be the line integral:

$$S = \oint_C (x + y)dx + (y - x)dy$$

- Calculate the line integral S directly
- Calculate the line integral using Greens Theorem

To compute this integral, C can firstly be rearranged, this is done by completing the square $(x-1)^2$:

$$C = x^2 + y^2 - 2x = 3 \Rightarrow x^2 - 2x + y^2 = 3 \Rightarrow x^2 - 2x + 1 + y^2 = 3 + 1 \Rightarrow (x-1)^2 + y^2 = 4$$

This is an equation for a cicle, as it takes the form $x^2 + y^2 = r^2$ with a center in $[1, 0]$ and $r = 2$. Because this is in the positive orientation, that means the rotation is counter-clockwise, therefore the standard parameterization is:

$$x^2 + y^2 = r^2 \qquad \Rightarrow \qquad x(t) = 1 + 2\ cos(t), \qquad y(t) = 2\ sin(t), \qquad 0 \leq t \leq 2\pi$$

The +1 in the x term is added because the circle is centered at $(1, 0)$ These can now be differentiated so that everything can be expressed in term of t:

$$\frac{dx}{dt} = -2\ sin(t) \qquad \frac{dy}{dt} = 2\ cos(t)$$

Everything can then be substituted into the line integral:

$$S = \oint_C (x + y)\ dx + (y - x)\ dy \Rightarrow \int_0^{2\pi} (1 + 2cos(t) + 2sin(t)) \cdot (-2sin(t))\ dt + (2sin(t) - 1 + cos(t)) \cdot (2cos(t))\ dt$$

This expression can be simplfied

$$= (-2)sin(t) - 4sin(t)\ cos(t) - 4sin^2(t) + 4sin(t)\ cos(t) - 2cos(t) - 4cos^2(t)$$

$$= (-2)(sin(t) - 2cos(t) - 4)(sin^2(t) + cos^2(t))$$

$$= (-2)(sin(t) + cos(t) - 4)$$

$$\Rightarrow \int_0^{2\pi} [(-2)(sin(t) + cos(t)) - 4]\, dt \Rightarrow (-2) \int sin(t)dt - 2 \int cos(t)dt - 4 \int dt$$

$$[2cos(t) - 2sin(t) - 4t]_0^{2\pi} \Rightarrow (2cos(2\pi) - 2sin(2\pi) - 4(2\pi)) - (2cos(0) - 2sin(0) - 4(0))$$

$$= 2 \cdot 1 - 0 - 8\pi - 2 \cdot 1 - 0 - 0$$

$$= -8\pi$$

This can also be evalutated using Greens Theorem:

$$\oint_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Here, here functions Q and P must firstly be identified from the given contour integral:

$$P = (x + y), \qquad Q = (y - x), \qquad \frac{\partial Q}{\partial x} = -1 \qquad \frac{\partial P}{\partial y} = 1$$

Into green's theorem:

$$\oint_C (x + y)dx + (y - x)dy = \iint_D (-1 - 1)\ dA \Rightarrow \iint_D -2\ dA$$

The same bounds of integration can be derived by shifting into polar coordinates, and since this is still a circle it takes the same form as before:

$$x = r\ cos(\theta), \qquad y = r\ sin(\theta),$$

And the bounds:

$$0 \leq r \leq 2, \qquad 0 \leq \theta \leq 2\pi$$

And:

$$dA = r\ dr\ d\theta$$

Evaluating the integral:

$$S = \int_0^{2\pi} \int_0^2 -2r\ dr\ d\theta$$

$$-2\int_0^2 r\ dr \Rightarrow -2\frac{1}{2}r^2 \Rightarrow \left[-(r^2)\right]_0^2 = (-(2)^2) - (-(0)^2) = -4$$

$$-4\int_0^{2\pi} d\theta \Rightarrow -4\theta \Rightarrow [-4\theta]_0^{2\pi} = (-4 \cdot 2\pi) - (-4 \cdot 0)$$

$$= -8\pi$$

5 Exercise 5

Let F be a vector field on $\mathbb{R}^3$ given by

$$F(x,y,z) = \frac{1}{3}(x^2, y^2, z^3)$$

And let C be the cube: $Cc = (x,y,z) \in \mathbb{R}^3 | 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.

- Calculate the flux out of the cube C directly
- Calculate the flux out of the cube C using Gauss Theorem (Divergence theorem)

Flux out of a cube, is the sum of the flux through all the flat surfaces of the cube. And can be calculated as

$$\Phi = \iint_S \mathbf{F} \cdot \mathbf{n} \; dS$$

The important detail here is that each surface of the cube, all six, must be calculated individually and then summed together.
Any of the six surfaces can be described by four points on any of the given two planes, xy, xz, yz. For the ZY-plane, it will be bounded by four points:

$$A = (0,0,0), \qquad B = (0,1,0), \qquad C = (0,1,1), \qquad D = (0,0,1)$$

This plane, which can also be called N_{x0} which is a side of the cube, is then the plane, which the flux will flow *outwards* from, meaning the normal vector to this plane must in the opposite direction of the cubes *depth*. This leads to the normal vector:

$$N_{x0} = (-1,0,0)$$

This rationality can then be applied to the rest of the cube as well.

$$N_{x1} = (1,0,0), \qquad N_{y0} = (0,-1,0), \qquad N_{y1} = (0,1,0), \qquad N_{z0} = (0,0,-1), \qquad N_{z1} = (0,0,1),$$

The flux through a single face(or surface) of the cube, is the bounded between 0 and 1 in any direction, and because one point is fixed in place and we make minute changes in direction in the two others, in the case of $x = 0$, $dS = dy\; dz$, describes the small steps we take. So the flux becomes:

$$\Phi_{x=0} = \int_0^1 \int_0^1 \frac{1}{3}(0,y,z) \cdot (-1,0,0) \; dy \; dz \Rightarrow \frac{1}{3}(0,y^2,z^2) \cdot (-1,0,0) \Rightarrow \frac{1}{3}\left(0 \cdot (-1) + y^2 \cdot 0 + z^2 \cdot 0\right) = 0$$

$$\Phi_{x=1} = \int_0^1 \int_0^1 \frac{1}{3}(1,y,z) \cdot (1,0,0) \; dy \; dz \Rightarrow \Phi_{x=1} = \int_0^1 \int_0^1 \frac{1}{3}(1,y^2,z^2) \cdot (1,0,0) \; dy \; dz$$

$$\Rightarrow \Phi_{x=1} = \int_0^1 \int_0^1 \frac{1}{3}\left(1 \cdot 1 + y^2 \cdot 0 + z^2 \cdot 0\right) \Rightarrow \frac{1}{3}\int_0^1 \int_0^1 \; dy \; dz \Rightarrow$$

$$\frac{1}{3}\int_0^1 [y]_0^1 \; dz \Rightarrow \frac{1}{3}\int_0^1 1 \; dz \Rightarrow \frac{1}{3}[z]_0^1 = \frac{1}{3}$$

$$\Phi_{y=0} = \int_0^1 \int_0^1 \frac{1}{3}(x,0,z) \cdot (0,-1,0) \; dx \; dz \Rightarrow \frac{1}{3}(x^2,0,z^2) \cdot (0,-1,0) \Rightarrow \frac{1}{3}\left(x^2 \cdot 0 + 0 \cdot (-1) + z^2 \cdot 0\right) = 0$$

$$\Phi_{y=1} = \int_0^1 \int_0^1 \frac{1}{3}(x,1,z) \cdot (0,1,0) \; dx \; dz \Rightarrow \Phi_{y=1} = \int_0^1 \int_0^1 \frac{1}{3}(x^2,1,z^2) \cdot (0,1,0) \; dx \; dz$$

$$\Rightarrow \Phi_{y=1} = \int_0^1 \int_0^1 \frac{1}{3}\left(x^2 \cdot 0 + 1 \cdot 1 + z^2 \cdot 0\right) \Rightarrow \frac{1}{3}\int_0^1 \int_0^1 \; dx \; dz \Rightarrow$$

$$\frac{1}{3}\int_0^1 [x]_0^1 \; dz \Rightarrow \frac{1}{3}\int_0^1 1dz \Rightarrow \frac{1}{3}[z]_0^1 = \frac{1}{3}$$

$$\Phi_{z=0} = \int_0^1\int_0^1 \frac{1}{3}(x,y,0)\cdot(0,0,-1)\; dx\; dy \Rightarrow \frac{1}{3}(x^2,y^2,0)\cdot(0,0,-1) \Rightarrow \frac{1}{3}\left(x^2\cdot 0 + y^2\cdot 0 + 0\cdot(-1)\right) = 0$$

$$\Phi_{z=1} = \int_0^1\int_0^1 \frac{1}{3}(x,y,1)\cdot(0,0,1)\; dx\; dy \Rightarrow \Phi_{z=1} = \int_0^1\int_0^1 \frac{1}{3}(x^2,1,z^2)\cdot(0,0,1)\; dx\; dy$$

$$\Rightarrow \Phi_{z=1} = \int_0^1\int_0^1 \frac{1}{3}\left(x^2\cdot 0 + y^2\cdot 0 + 1\cdot 1\right) \Rightarrow \frac{1}{3}\int_0^1\int_0^1 \; dx\; dy \Rightarrow$$

$$\frac{1}{3}\int_0^1 [x]_0^1 \; dy \Rightarrow \frac{1}{3}\int_0^1 1dy \Rightarrow \frac{1}{3}[y]_0^1 = \frac{1}{3}$$

This yields the final calculation for the flux out of the cube as:

$$\Phi = \Phi_{x=0} + \Phi_{x=1} + \Phi_{y=0} + \Phi_{y=1} + \Phi_{z=0} + \Phi_{z=1} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

Alternatively, this can be calculated using the divergence (or Gauss theorem). It essentially states that we can, by deriving the divergence inside the enclosed space of the cube, see how much our vector fields behave like a source or a sink at each point inside it, and then sum all those point up. Thus it takes the form of:

$$\iiint_V \nabla\cdot\mathbf{F}\; dV$$

Starting from the vector field $\mathbf{F}$

$$F(x,y,z) = \frac{1}{3}(x^2,y^2,z^2)$$

The divergence can be found, as the sum of all the partial derivatives

$$\nabla\cdot\mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = \frac{\partial}{\partial x}\left(\frac{x^2}{3}\right) + \frac{\partial}{\partial y}\left(\frac{y^2}{3}\right) + \frac{\partial}{\partial z}\left(\frac{z^2}{3}\right) \Rightarrow \frac{2x}{3} + \frac{2y}{3} + \frac{2z}{3} \Rightarrow \frac{2}{3}(x+y+z)$$

The bounds of integration are now the three dimensional volume of the cube from 0 to 1.

$$\Phi = \int_0^1\int_0^1\int_0^1 \frac{2}{3}(x+y+z)\; dx\; dy\; dz$$

$$\Rightarrow \int_0^1 (x+y+z)dx \Rightarrow \frac{2}{3}\left[\frac{x^2}{2} + xy + xz\right]_0^1 = \frac{2}{3}\left(\frac{1}{2} + (1)y + (1)z\right) = \frac{2}{3}\left(\frac{1}{2} + y + z\right)$$

$$\Rightarrow \frac{2}{3}\int_0^1 \left(\frac{1}{2} + y + z\right) \Rightarrow \frac{2}{3}\left[\frac{y}{2} + \frac{y^2}{2} + yz\right]_0^1 = \frac{2}{3}(1+z)$$

$$\Rightarrow \frac{2}{3}\int_0^1 (1+z) = \frac{2}{3}\left[z + \frac{z^2}{2}\right]_0^1 = \frac{2}{3}\left(1 + \frac{1}{2}\right) = \frac{2}{3}\cdot\frac{3}{2} = 1$$